Analysis of Clustered Feeding Behavior: From Exponential Decay to Conditional Probabilities

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1 Introduction

In studying feeding behavior, one important question is how animals cluster their feeding events: do they consume food in short bursts (snacks), moderate sessions (meals), or extended bouts (feasts)? We explore this question using a two-part approach:

1. **Exponential Decay Model:** We first examine the overall frequency distribution of cluster sizes, hypothesizing that larger clusters will be increasingly rare, describable by an exponential decay.
2. **Conditional Probability Analysis:** Next, we compute the likelihood that a feeding cluster of size x grows to size $x + 1$. We also calculate confidence intervals to gauge the statistical certainty of these transition probabilities.

Our final goal is to propose boundaries for categorizing feeding events into:

- *Snack*: 1 pellet,
- *Meal*: 2–5 pellets,
- *Feast*: more than 5 pellets.

2 Exponential Decay Analysis

2.1 Model and Fitting Procedure

Let x represent the cluster size (number of pellets consumed in a single feeding event) and y represent the observed frequency (or average count) of clusters of size x. We propose the exponential decay model:

$$y = a\,e^{-bx},$$

where:

- a is a scale parameter, corresponding to the extrapolated frequency at $x = 0$,

- b is the decay constant, controlling how quickly the frequency diminishes as x increases.

We fit this model to the empirical cluster-size distribution (typically restricting x to a reasonable range, e.g., 1 to 10) using non-linear least squares.

2.2 Results and Interpretation

A sample fit yielded:

$$\hat{a} \approx 139.55, \quad \hat{b} \approx 0.28, \quad R^2 \approx 0.9668.$$

The high R^2 indicates that the exponential function explains approximately 96.68% of the variance in the cluster-size data. Such a decay pattern suggests that although small-to-moderate cluster sizes are common, very large clusters are increasingly rare.

3 Conditional Probability of Cluster Growth

3.1 Definition of $P(x+1 \mid x)$

To further characterize feeding patterns, we compute the probability that a cluster of size x will grow by one pellet to size $x+1$:

$$P(x+1 \mid x) \; = \; \frac{C(x+1)}{C(x)},$$

where $C(x)$ is the count of observed clusters of size x. If $C(x) = 0$, then $P(x+1 \mid x) = 0$ by definition.

3.2 Empirical Estimates

For mice in a non-restricted (NR) feeding phase, limited to those meeting an "Order 1" criterion, we observed:

$$\begin{aligned}
P(2 \mid 1) &= 0.6293, \\
P(3 \mid 2) &= 0.9747, \\
P(4 \mid 3) &= 0.7913, \\
P(5 \mid 4) &= 0.7897, \\
P(6 \mid 5) &= 0.5502, \\
P(7 \mid 6) &= 0.7817, \\
P(8 \mid 7) &= 0.5838, \\
P(9 \mid 8) &= 0.6609, \\
P(10 \mid 9) &= 0.6316.
\end{aligned}$$

Notably, once a cluster reaches size 2, it is very likely (≈ 0.97) to continue to size 3. Conversely, the step from 5 to 6 shows a lower probability (≈ 0.55), indicating a natural "choke point" where many clusters stop growing.

4 Confidence Intervals for $P(x+1 \mid x)$

4.1 Wilson Score Intervals

To assess the uncertainty of each conditional probability estimate, we compute 95% confidence intervals (CIs) using the Wilson score method. For each cluster size x:

$$\text{trials} = C(x), \quad \text{successes} = C(x+1), \quad \hat{p} = \frac{\text{successes}}{\text{trials}}.$$

The Wilson method provides stable CI bounds, particularly suitable when proportions are near 0 or 1, or when sample sizes vary.

4.2 Results

Table 1 shows the observed conditional probabilities along with their 95% Wilson CIs:

Table 1: Observed Conditional Probabilities and 95% Confidence Intervals

Current Size	$\mathbf{P(x+1 \mid x)}$	**Lower 95% CI**	**Upper 95% CI**
1	0.6293	0.6015	0.6562
2	0.9747	0.9609	0.9838
3	0.7913	0.7604	0.8191
4	0.7897	0.7546	0.8209
5	0.5502	0.5044	0.5952
6	0.7817	0.7267	0.8283
7	0.5838	0.5140	0.6503
8	0.6609	0.5704	0.7409
9	0.6316	0.0000	0.0481[1]

5 Classification into Snack, Meal, and Feast

5.1 Rationale

Combining the exponential decay insights (where large cluster sizes are rare) with conditional probability trends (which drop appreciably around size 5 or 6), we propose:

$$\begin{aligned} \text{Snack} &:= 1 \text{ pellet}, \\ \text{Meal} &:= 2 \le x \le 5 \text{ pellets}, \\ \text{Feast} &:= x > 5 \text{ pellets}. \end{aligned}$$

This distinction is justified by:

- **High growth probability** below size 5 (e.g., going from 2 to 3, or 3 to 4).

[1] Sparse data at cluster size 9 led to an extreme or imprecise CI bound.

- **Moderate or low growth probability** at or above size 5–6.
- **Exponential decay** indicating fewer events above size 5, consistent with a biological boundary separating regular "meals" from extended "feasts."

6 Conclusion

Through an integrated approach—fitting an exponential decay to cluster size frequencies and computing conditional probabilities (plus confidence intervals)—we gain detailed insights into how feeding events grow.

- *Small clusters* (especially 1 pellet) rarely add more pellets, making them good "snack" candidates.
- *Intermediate clusters* (2–5 pellets) tend to continue with moderate-to-high probability, supporting their classification as "meals."
- *Large clusters* (> 5 pellets) become noticeably less common (exponential decay) and may represent distinct "feasts."

This data-driven framework therefore offers a refined method to categorize feeding events into biologically meaningful types.